

Tiny Rule-Based Expert System

Run: python P_rule_based_expert_system.py

Program

Tiny Rule-Based Expert System

Domain: Animal identification (forward-chaining inference engine)

```
rules = [
    {"if": {"has_hair"}, "then": "is_mammal"},
    {"if": {"gives_milk"}, "then": "is_mammal"},
    {"if": {"has_feathers"}, "then": "is_bird"},
    {"if": {"flies", "lays_eggs"}, "then": "is_bird"},
    {"if": {"is_mammal", "eats_meat"}, "then": "is_carnivore"},
    {"if": {"is_mammal", "has_pointed_teeth", "has_claws", "eyes_forward"}, "then": "is_carnivore"},
    {"if": {"is_mammal", "has_hooves"}, "then": "is_ungulate"},
    {"if": {"is_mammal", "chews_cud"}, "then": "is_ungulate"},
    {"if": {"is_carnivore", "has_tawny_color", "has_dark_spots"}, "then": "animal_is_cheetah"},
    {"if": {"is_carnivore", "has_tawny_color", "has_black_stripes"}, "then": "animal_is_tiger"},
    {"if": {"is_ungulate", "has_long_neck", "has_long_legs"}, "then": "animal_is_giraffe"},
    {"if": {"is_ungulate", "has_black_stripes"}, "then": "animal_is_zebra"},
    {"if": {"is_bird", "does_not_fly", "has_long_neck", "has_long_legs", "is_black_and_white"}, "then": "animal_is_ostrich"},
    {"if": {"is_bird", "swims", "does_not_fly", "is_black_and_white"}, "then": "animal_is_penguin"},
    {"if": {"is_bird", "flies_well"}, "then": "animal_is_albatross"},
]

def infer(facts):
    """Forward chaining: keep applying rules until no new fact is derived."""
    facts = set(facts)
    fired = []
    changed = True
    while changed:
        changed = False
        for rule in rules:
            if rule["if"].issubset(facts) and rule["then"] not in facts:
                facts.add(rule["then"])
                fired.append(rule)
                changed = True
    return facts, fired

def diagnose(name, given_facts):
    print("=" * 55)
    print(f"Case: {name}")
    print(f"Given facts: {sorted(given_facts)}")
    final_facts, fired_rules = infer(given_facts)
    print("\nRules fired, in order:")
    if not fired_rules:
        print(" (none)")
    for i, r in enumerate(fired_rules, 1):
        print(f" {i}. IF {sorted(r['if'])} THEN {r['then']}")
    conclusion = [f for f in final_facts if f.startswith("animal_is_")]
    print("\nConclusion:", conclusion[0].replace("animal_is_", "").upper()
          if conclusion else "NOT ENOUGH INFORMATION TO IDENTIFY THE ANIMAL")
    print()

if __name__ == "__main__":
    diagnose("Case 1", {
        "has_hair", "eats_meat",
        "has_tawny_color", "has_black_stripes"
    })

    diagnose("Case 2", {
        "has_hair", "has_hooves", "chews_cud",
        "has_long_neck", "has_long_legs"
    })

    diagnose("Case 3", {
        "has_feathers", "swims", "does_not_fly", "is_black_and_white"
    })

    diagnose("Case 4", {
        "has_hair", "has_pointed_teeth", "has_claws", "eyes_forward",
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    "has_tawny_color", "has_dark_spots"  
  })  
  
  diagnose("Case 5 (insufficient facts)", {  
    "has_hair", "eats_meat"  
  })  
}
```

Output

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Case: Case 1

Given facts: ['eats_meat', 'has_black_stripes', 'has_hair', 'has_tawny_color']

Rules fired, in order:

1. IF ['has_hair'] THEN is_mammal
2. IF ['eats_meat', 'is_mammal'] THEN is_carnivore
3. IF ['has_black_stripes', 'has_tawny_color', 'is_carnivore'] THEN animal_is_tiger

Conclusion: TIGER

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Case: Case 2

Given facts: ['chews_cud', 'has_hair', 'has_hooves', 'has_long_legs', 'has_long_neck']

Rules fired, in order:

1. IF ['has_hair'] THEN is_mammal
2. IF ['has_hooves', 'is_mammal'] THEN is_ungulate
3. IF ['has_long_legs', 'has_long_neck', 'is_ungulate'] THEN animal_is_giraffe

Conclusion: GIRAFFE

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Case: Case 3

Given facts: ['does_not_fly', 'has_feathers', 'is_black_and_white', 'swims']

Rules fired, in order:

1. IF ['has_feathers'] THEN is_bird
2. IF ['does_not_fly', 'is_bird', 'is_black_and_white', 'swims'] THEN animal_is_penguin

Conclusion: PENGUIN

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Case: Case 4

Given facts: ['eyes_forward', 'has_claws', 'has_dark_spots', 'has_hair', 'has_pointed_teeth', 'has_tawny_color']

Rules fired, in order:

1. IF ['has_hair'] THEN is_mammal
2. IF ['eyes_forward', 'has_claws', 'has_pointed_teeth', 'is_mammal'] THEN is_carnivore
3. IF ['has_dark_spots', 'has_tawny_color', 'is_carnivore'] THEN animal_is_cheetah

Conclusion: CHEETAH

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Case: Case 5 (insufficient facts)

Given facts: ['eats_meat', 'has_hair']

Rules fired, in order:

1. IF ['has_hair'] THEN is_mammal
2. IF ['eats_meat', 'is_mammal'] THEN is_carnivore

Conclusion: NOT ENOUGH INFORMATION TO IDENTIFY THE ANIMAL